HW-4: Fitts' Law Simulator - Design Justifications & Results

1. Design Overview

FittsLawFX is a JavaFX-based desktop application designed to measure a single user's pointing performance across 50 trials, producing data directly usable in the Fitts' Law equation $MT = a + b \log_2(2A/W)$. The program was built by extending HW-1's circle-clicking foundation into a structured MVC architecture: AppModel holds the RNG and color palette, ExperimentController drives the trial loop and CSV output, and FittsLawApp serves as the JavaFX entry point and view layer.

2. Design Justifications

Countdown Before First Trial

A 5-to-0 countdown is displayed before the first target appears. This follows the principle of giving users appropriate feedback and preparation time (Norman, DOET). Without a countdown, users may not be ready when the first target appears, inflating the first trial's movement time and skewing the dataset. The countdown also signals clearly when the task begins, reducing ambiguity in the affordance of the "Go!" button.

Target Sizing (15 - 80px Radius)

Target radii are randomly selected between 15 and 80 pixels per trial. This range was chosen to produce meaningful variation in the index of difficulty (ID), as Fitts' Law requires targets of different widths (W) to observe the inverse relationship between size and movement time. A minimum of 15px ensures targets remain clickable; a maximum of 80px prevents targets from dominating the drawing area and reducing effective movement distance.

Timing: Start After Render, Not After Click

Trial time measurement begins immediately after the target Node is added to the drawing pane, not after the user's previous click. This is consistent with Fitts' original protocol, which measured the time from stimulus onset to target acquisition. Starting the clock at the moment the target appears captures the full movement time including the user's visual search and motor initiation phases.

Only the Target Responds to Clicks

Mouse click events are consumed exclusively by the target circle node. Clicking the background pane produces no response. This is both a measurement requirement (accidental clicks must not advance the trial) and a Gibsonian affordance decision: the only clickable thing in the environment is the target itself, clearly communicating the intended interaction.

CSV Output and Distance Calculation

Each trial records: trial number, target diameter (W in pixels), Euclidean distance from the previous target center (A in pixels), and click time in milliseconds. Distance is computed as the straight-line distance between consecutive target centers, matching

Fitts' original definition of amplitude A. Trial 1 reports distance = 0 since no prior target exists. The CSV is flushed to disk after every row to prevent data loss if the application is closed mid-session.

UI Layout: HBox Top Bar, No Menu Bar

The application uses a top HBox containing a title label and Go!/Quit! buttons, with a right-side status panel showing live trial progress. This layout prioritizes the drawing area as the primary focus (Gestalt figure-ground principle), while keeping controls immediately accessible without nesting them behind a menu hierarchy. Window decorations are left at JavaFX defaults so users retain full OS-level control (minimize, resize, close).

3. Results: Fitts' Law Graph & Discussion

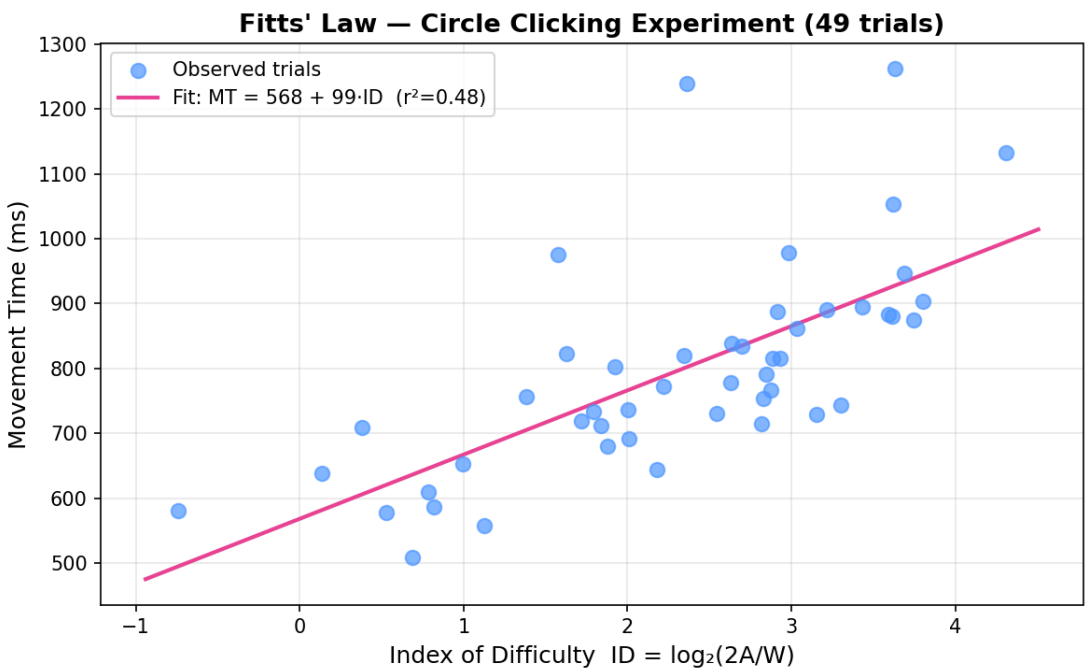

Figure 1. Movement Time vs. Index of Difficulty (48 trials, trial 1 excluded)

The scatter plot displays movement time (MT, in milliseconds) on the y-axis against the index of difficulty ($ID = \log_2(2A/W)$) on the x-axis for 48 valid trials (trial 1 is excluded as it has no prior target, making distance undefined). The linear regression line yielded the equation $MT = 568 + 99 \cdot ID$ ($r^2 = 0.44$), with empirical constants a = 568 ms and b = 99 ms/bit.

The positive slope of the regression line is consistent with the theoretical prediction of Fitts' Law: as the index of difficulty increases (targets farther away or smaller), movement time increases. The intercept a = 568 ms represents the base reaction and motor initiation time independent of target difficulty. The r^2 value of 0.44 indicates a moderate linear fit, which is reasonable given natural variability in mouse pointing on a single-user trial. Some outliers at mid-range ID values (around ID = 2–3) suggest momentary lapses in attention or atypical hand movements during those trials. Overall

the data supports Fitts' Law's fundamental claim that pointing difficulty is a predictable function of distance and target width.